

BALACHANDRAKUMAR B

AI Engineer | Machine Learning Engineer | Computer Vision Engineer

Nagercoil, Tamil Nadu, India | +91 91505 21955 | balachandrakumar001b@gmail.com | LinkedIn | GitHub

PROFESSIONAL SUMMARY

B.E. Computer Science and Engineering graduate with hands-on experience in developing Artificial Intelligence, Machine Learning, Deep Learning, and Computer Vision applications. Skilled in Python, Scikit-learn, TensorFlow/PyTorch concepts, CNN, GRU, OpenCV, and AI model development workflows including data preprocessing, model training, evaluation, and deployment. Built real-world AI solutions including a deepfake detection and anti-spoofing system and an intelligent PDF automation platform. Seeking an entry-level AI Engineer / Machine Learning Engineer role to apply skills in deep learning, computer vision, and Python-based AI application development.

TECHNICAL SKILLS

Programming: Python, SQL, Java (Basic)

AI / Machine Learning: Machine Learning, Deep Learning, Supervised Learning, Classification, Regression, Feature Engineering, Model Training, Model Evaluation

Deep Learning: CNN, GRU, Neural Networks, EfficientNet-B0, Transfer Learning

Computer Vision: OpenCV, MediaPipe, Image Processing, Face Detection, Face Recognition, Deepfake Detection

Data Science: NumPy, Pandas, Matplotlib, EDA, Data Cleaning, Data Preprocessing

AI Deployment: FastAPI, REST API, Streamlit, Model Deployment

Tools: Git, GitHub, Jupyter Notebook, Google Colab, VS Code

PROJECTS

AI Security Suite — Deepfake Detection & Anti-Spoofing using Deep Learning

Final Year Project

Python, OpenCV, MediaPipe, CNN, GRU, EfficientNet-B0, FFT, Streamlit

- Developed a deep learning based REAL/FAKE classification system for image, video, and webcam inputs using CNN, GRU, and EfficientNet-B0 models.
- Integrated MediaPipe facial landmark tracking and FFT frequency analysis to improve deepfake detection and liveness verification.
- Implemented real-time webcam inference using Streamlit for identity verification and anti-spoofing applications.

PDFPro — Intelligent PDF Processing Toolkit

Personal Project

Python, FastAPI, PyMuPDF, OpenPyXL

- Developed 50+ PDF automation tools including merge, split, compression, OCR extraction, encryption, and document conversion.
 - Built a FastAPI-based backend with offline processing capability and optimized document workflows.
 - Improved usability with automated processing features, file handling, and interactive application interface.
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Machine Learning Project — VCodez Internship

Python, Pandas, NumPy, Scikit-learn

- Developed and evaluated machine learning classification and regression models using Scikit-learn.
 - Performed data preprocessing, feature engineering, model training, and hyperparameter tuning.
 - Achieved 89% model accuracy through evaluation and optimization techniques.
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INTERNSHIP EXPERIENCE

Machine Learning Intern — VCodez

May 2025 – Aug 2025

- Worked on complete ML workflow including data preprocessing, feature engineering, model development, and evaluation.
- Implemented machine learning algorithms using Python and Scikit-learn for practical AI applications.

Web Development Intern — AK Infopark

Apr 2024

- Developed responsive web interfaces using HTML, CSS, and JavaScript for client-based projects.
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EDUCATION

B.E. Computer Science and Engineering

Jayamatha Engineering College (Anna University)

2022 – 2026 | CGPA: 8.0 / 10

CERTIFICATIONS

- Machine Learning Internship Certificate — VCodez
- Machine Learning Training Certificate — Besant Technologies
- Web Development Internship Certificate — AK Infopark